

Professional Summary

Mechanical engineering student seeking a mechanical design role to apply skills in CAD (Fusion 360, AutoCAD) and CAE (ABAQUS, ANSYS) that will help contribute to the design, validation and optimization of innovative products

Key Skills

Machine Design | Fusion 360 | Design for Manufacturing | Robotic Kinematics | Rapid Prototyping | ANSYS | ABAQUS | Manufacturing Processes | AutoCAD | Mechanics of Solids | Fluid Mechanics | Heat Transfer | Material Science | Vibrations and Control | Mechanisms and Machines | Python | Machine Learning | AI for Robotics |

Projects

Gripper System

- Engineered a high-torque robotic gripper using Fusion 360 using the 4 - bar linkage system, optimized for 3D printing with integrated bosses for structural integrity.
- Performed torque requirement analysis to select the MG958 servo, implementing a 3:1 gear ratio to maximize gripping force while maintaining motor efficiency.
- Sourced and integrated precision hardware, including bearings (MR 105), fasteners (nuts, bolts and washers, and custom-treaded gripper fingers to enhance mechanical advantage and friction to improve grip.

Dry Wall Taping End Effector

- Designed and modeled a robotic drywall taping end effector (payload capacity 8kg) in Fusion 360, reducing total assembly weight to 2.78kg (fully loaded) by selecting lightweight materials like HDPE and Acetal
- Developed an automated drywall finisher featuring a belt-driven tape feeder, a compound reservoir, and a corrosion-resistant scraper for consistent seam finishing.
- Performed FMEA to mitigate tape tearing and scraper wear; specified CNC and laser-cut components and established maintenance protocols for long-term durability.

Rack and Pinion Gear Mechanism

- Designed a rack and pinion gripper to lift a 2kg object while strictly adhering to a weight constraint of 100g. Optimized the final assembly weight, including the motor, to 97.03g.
- Performed force analysis to determine a 28.03N clamping requirement ($FS=2$, frictional coefficient = 0.7); selected EMAX ES3004 servo (3.5 kg-cm) to exceed the calculated 3.265 kg-cm torque threshold.

Education

Birla Institute of Technology and Science Pilani, Goa | Mechanical Engineering | CGPA: 8.70 | Oct 2022 – June 2026

Additional Training & Certifications

Supervised Machine Learning: Regression and Classification | 2023

Awards

BITSAA UK Scholarship | 2024

